

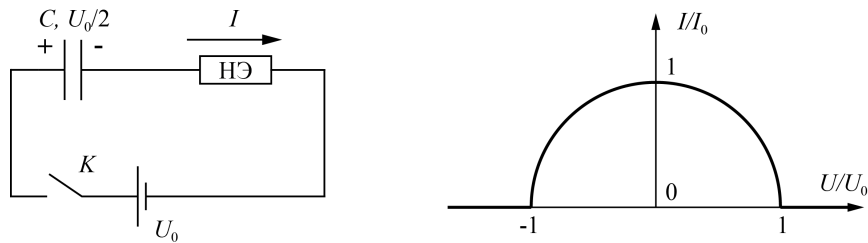
# Nonlinear element

W21 (2021) — English translation

The electrical circuit consists of a constant voltage source  $U_0$ , a capacitor of capacitance  $C$ , a nonlinear element NE, and a switch  $K$ . The current-voltage characteristic of the nonlinear element is as follows:

$$\left(\frac{I}{I_0}\right)^2 + \left(\frac{U}{U_0}\right)^2 = 1$$

for  $I > 0$  and  $I = 0$  for  $|U| \geq U_0$  (see figure). The value  $I_0$  is known. At the initial moment, the voltage across the capacitor is  $U_0/2$ . The switch  $K$  is closed.



**Figure 1:** Figure 1.

|           |                                                                                                                                               |             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>A1</b> | Find the maximum power $P_1$ released on the nonlinear element.                                                                               | <b>2.00</b> |
| <b>A2</b> | Find the maximum rate of change of energy of the capacitor $P_2$ and the time after closing the switch $\Delta t$ after which it is achieved. | <b>8.00</b> |

Source: <https://pho.rs/p/80>